

# TU VU

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## APPOINTMENTS

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### Virginia Tech

Fall 2024 — *present*

Assistant Professor, Computer Science

Research Interests: *natural language processing & machine learning*

### Google

Faculty Researcher

Spring 2025 — *present*

Google DeepMind Research Scientist

2023 — 2024

## EDUCATION

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### University of Massachusetts, Amherst

2016 — 2023

M.S/PH.D. in Computer Science

Advisor: [Mohit Iyyer](#)

2018 — 2023

Thesis committee: [Mohit Iyyer](#), [Subhransu Maji](#), [Hamed Zamani](#), [Thang Luong](#), [Colin Raffel](#)

### Vietnam National University, Hanoi

2009 — 2013

B.S. Honors Program in Computer Science

Highest distinction (class rank: 1/100)

## PROFESSIONAL EXPERIENCE

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### Google DeepMind

Fall 2022 — Spring 2023

Student Researcher *with* [Thang Luong](#) & [Quoc Le](#)

### Google DeepMind

Summer 2022

Research Intern *with* [Thibault Sellam](#) & [Elizabeth Clark](#)

### Google DeepMind

Winter 2021 — Spring 2022

Student Researcher *with* [Noah Constant](#)

### Google DeepMind

Summer 2021 — Fall 2021

Research Intern & Student Researcher *with* [Daniel Cer](#) & [Noah Constant](#)

### Google DeepMind

Winter 2020 — Spring 2021

Student Researcher *with* [Thang Luong](#) & [Quoc Le](#)

### Google DeepMind

Summer 2020

Research Intern *with* [Grady Simon](#) & [Zi Yang](#) & [Nan Hua](#)

### Microsoft Research

Summer 2019

Research Intern *with* [Tong Wang](#) & [Tsendsuren Munkhdalai](#) & [Adam Trischler](#)

\*: equal contribution

For an up-to-date list of my research papers, please see my [Google Scholar](#) profile.

SealQA: Raising the Bar for Reasoning in Search-Augmented Language Models

Thinh Pham, Nguyen Nguyen, Pratibha Zunjare, Weiyuan Chen, Yu-Min Tseng, **Tu Vu**  
**arXiv**

*// Our benchmark dataset has been used by Google's Gemini, DeepSeek, and Kimi*

Efficient Model Development through Fine-tuning Transfer

Pin-Jie Lin, Rishab Balasubramanian, Fengyuan Liu, Nikhil Kandpal, **Tu Vu**

**EMNLP 2025**

What Matters for Model Merging at Scale?

Prateek Yadav, **Tu Vu**, Jonathan Lai, Alexandra Chronopoulou, Manaal Faruqui, Mohit Bansal, Tsendsuren Munkhdalai

**TMLR 2025**

Foundational Autoraters: Taming Large Language Models for Better Automatic Evaluation

**Tu Vu\***, Kalpesh Krishna\*, Salaheddin Alzubi, Chris Tar, Manaal Faruqui, Yun-Hsuan Sung

**EMNLP 2024**

*// The top-performing generative model on RewardBench as of July 15, 2024, trained only on publicly available data*

Gemini: A Family of Highly Capable Multimodal Models

Google Gemini Team: Rohan Anil, Rohan Anil, Sebastian Borgeaud, Yonghui Wu, Jean-Baptiste Alayrac, Jiahui Yu, Radu Soricut, Johan Schalkwyk, Andrew Dai, Anja Hauth, and others including

**Tu Vu**

**Technical report 2023**

*// Google AI Blog*

FreshLLMs: Refreshing Large Language Models with Search Engine Augmentation

**Tu Vu**, Mohit Iyyer, Xuezhi Wang, Noah Constant, Jerry Wei, Jason Wei, Chris Tar, Yun-Hsuan Sung, Denny Zhou, Quoc Le, and Thang Luong

**ACL 2024 Findings**

*// Our dataset and method have inspired or been used for the development of Google's Gemini, Perplexity.AI's Online LLMs, You.com, and Contextual AI's RAG 2.0*

The Flan Collection: Designing Data and Methods for Effective Instruction Tuning

Shayne Longpre, Le Hou, **Tu Vu**, Albert Webson, Hyung Won Chung, Yi Tay, Denny Zhou, Quoc Le, Barret Zoph, Jason Wei, and Adam Roberts

**ICML 2023**

*// Google Research Blog*

Mixture-of-experts meets instruction tuning: A winning combination for large language models

Sheng Shen, Le Hou, Yanqi Zhou, Nan Du, Shayne Longpre, Jason Wei, Hyung Won Chung, Barret Zoph, William Fedus, Xinyun Chen, **Tu Vu**, Yuexin Wu, Wuyang Chen, Albert Webson, Yunxuan Li, Vincent Zhao, Hongkun Yu, Kurt Keutzer, Trevor Darrell, and Denny Zhou

**ICLR 2024**

SPoT: Better Frozen Model Adaptation through Soft Prompt Transfer

**Tu Vu**, Brian Lester, Noah Constant, Rami Al-Rfou, and Daniel Cer

## ACL 2022

Overcoming Catastrophic Forgetting in Zero-Shot Cross-Lingual Generation

Tu Vu, Aditya Barua, Brian Lester, Daniel Cer, Mohit Iyyer, and Noah Constant

EMNLP 2022

STraTA: Self-Training with Task Augmentation for Better Few-shot Learning

Tu Vu, Thang Luong, Quoc Le, Grady Simon, and Mohit Iyyer

EMNLP 2021

Exploring and Predicting Transferability across NLP Tasks

Tu Vu, Tong Wang, Tsendsuren Munkhdalai, Alessandro Sordani, Adam Trischler, Andrew Mattarella-Micke, Subhransu Maji, and Mohit Iyyer

EMNLP 2020

## FUNDING & RESEARCH SUPPORT

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|                                                |      |
|------------------------------------------------|------|
| Google DeepMind Research Gift Award (\$30,000) | 2025 |
| PI: Tu Vu                                      |      |
| Amazon - VT Faculty Research Award (\$130,000) | 2025 |
| PI: Tu Vu                                      |      |
| Google AI Studio Tier 1                        | 2025 |
| PI: Tu Vu                                      |      |
| VT New Faculty Mentoring Grant (\$1,500)       | 2025 |
| PI: Tu Vu                                      |      |
| Adobe Research Gift (\$5,000)                  | 2024 |
| PI: Tu Vu                                      |      |

## ADVISING

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### PHD ADVISEES:

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|----------------------------------------------------------|----------------------------|
| Weiyuan Chen, 1 <sup>st</sup> year PhD student           | Fall 2025 — <i>present</i> |
| Yu-Min Tseng, 1 <sup>st</sup> year PhD student           | Fall 2025 — <i>present</i> |
| Jing Chen, 1 <sup>st</sup> year PhD student              | Fall 2025 — <i>present</i> |
| Rishab Balasubramanian, 2 <sup>nd</sup> year PhD student | Fall 2024 — <i>present</i> |
| Thinh Pham, 2 <sup>nd</sup> year PhD student             | Fall 2024 — <i>present</i> |
| Quyet Do, 2 <sup>nd</sup> year PhD student               | Fall 2024 — <i>present</i> |
| Pin-Jie Lin, 2 <sup>nd</sup> year PhD student            | Fall 2024 — <i>present</i> |

### MASTERS ADVISEES:

|                                                  |                              |
|--------------------------------------------------|------------------------------|
| Rituraj Sharma, 1 <sup>st</sup> year MS student  | Spring 2025 — <i>present</i> |
| Noah Provenzano, 2 <sup>nd</sup> year MS student | Spring 2025 — <i>present</i> |

### UNDERGRAD ADVISEES:

|                               |                              |
|-------------------------------|------------------------------|
| Nguyen Nguyen, junior student | Spring 2025 — <i>present</i> |
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## OTHERS:

|                                                           |                           |
|-----------------------------------------------------------|---------------------------|
| Zhenting Qi, Student Researcher @ <b>Google</b>           | Summer — Fall 2025        |
| Prateek Yadav, Research Intern @ <b>Google DeepMind</b>   | Summer 2024 — Spring 2025 |
| Simeng Han, Student Researcher @ <b>Google DeepMind</b>   | Summer 2024 — Spring 2025 |
| Salaheddin Alzubi, Masters student @ <b>UMass Amherst</b> | Fall 2022 — Spring 2023   |
| Dheeraj Mekala, PhD student @ <b>UCSD</b>                 | Spring — Summer 2022      |

## TEACHING

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|                                      |             |
|--------------------------------------|-------------|
| CS 4804: Introduction to AI          | Fall 2025   |
| CS 5624: Natural Language Processing | Spring 2025 |

## RECENT INVITED TALKS & LECTURES

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|                                                                                                                                                  |               |
|--------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| Attention Mechanisms and Transformers<br>Two lectures @ <b>The New Turing Institute</b>                                                          | August 2025   |
| SealQA: Raising the Bar for Reasoning in Search-Augmented Language Models<br><b>Open AGI Symposium @ UC Berkeley</b> , hosted by <b>Sentient</b> | August 2025   |
| Multimodal LLMs & Test-time Scaling<br>Lecture @ <b>The New Turing Institute</b>                                                                 | June 2025     |
| Efficient Model Development in the Era of Large Language Models<br><b>Qualcomm</b>                                                               | November 2024 |
| Efficient Model Development in the Era of Large Language Models<br><b>Mila / McGill NLP Seminar</b>                                              | October 2024  |
| Efficient Adaptation of Large Language Models<br>Graph Neural Networks Reading Group, <b>Google</b>                                              | November 2023 |
| Effective and Efficient Transfer Learning<br>in the Era of Large Language Models<br><b>Faculty job talk</b>                                      | Spring 2023   |
| Overcoming Catastrophic Forgetting<br>in Zero-Shot Cross-Lingual Generation<br>Parameter Efficient Tuning Methods Sync, <b>Google</b>            | October 2022  |
| Transfer Learning with Large-scale Language Models<br>Lecture @ <b>The New Turing Institute</b>                                                  | August 2022   |
| The Appeal of Parameter-efficient Transfer Learning<br>Natural Language Accelerated Team, <b>Google</b>                                          | June 2022     |
| SPoT: Better Frozen Model Adaptation through Soft Prompt Transfer<br>Parameter Efficient Tuning Methods Sync, <b>Google</b>                      | December 2021 |

**Area Chair** for ACL 2025, 2024; EMNLP 2025, 2024, NAACL 2025; COLING 2025

**Session Chair** (Machine Learning for NLP) @ EMNLP 2024

**Program Committee/Reviewer** for NATURE 2025, NEURIPS 2025, 2024; ICLR 2025; ICML 2025; TL4NLP@NEURIPS 2022; MCDL@ICLR 2025; COLM 2025, 2024; AAAI 2023; ACL 2022, 2021, 2020, 2019; EMNLP 2022, 2021; NAACL 2022, 2021; COLING 2020; CoNLL 2019; INLG 2020, 2019

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## SELECTED MEDIA

|                                                                                        |          |
|----------------------------------------------------------------------------------------|----------|
| GEMINI: <a href="#">Google AI Blog</a>                                                 | 2023     |
| FRESHLLMs: <a href="#">ZDNET</a>                                                       | 2023     |
| THE FLAN COLLECTION: <a href="#">Google Research Blog</a>                              | 2023     |
| SPoT: <a href="#">Headlines of Google AI's Natural Language Accelerated Newsletter</a> | Q1, 2022 |

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## SELECTED AWARDS & HONORS

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|-------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Google Student Researchships                                                                                                                          | 2020 — 2023 |
| UMass Amherst Graduate Assistantships                                                                                                                 | 2016 — 2023 |
| Honda Y-E-S Award for young engineers and scientists, Vietnam<br><i>// in the top 10 nationally</i>                                                   | 2013        |
| Outstanding Academic and Co-curricular Achievements, Vietnam National University                                                                      | 2013        |
| Prominent Young Figure Award, Vietnam National University                                                                                             | 2010 & 2012 |
| First Runner-up Prize, International Programming Contest, Japan<br><i>// ranked 2<sup>nd</sup> among 64 teams internationally</i>                     | 2011        |
| Outstanding Young Talent of the Capital City, Vietnam<br><i>// in the top 100 most outstanding young talents selected from a wide range of fields</i> | 2010        |
| Champion Prize, National Mathematical Olympiad, Vietnam<br><i>// ranked 1<sup>st</sup> among more than 600 contestants nationally</i>                 | 2010        |
| A number of prizes in National/International Olympiads<br>(in both Mathematics and Informatics)                                                       | 2009 — 2013 |
| A number of academic scholarships for undergraduate students                                                                                          | 2009 — 2013 |

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## PATENTS

\*: original inventor

[Frozen Model Adaptation Through Soft Prompt Transfer](#)

**Tu Vu\***, [Daniel Cer](#), [Noah Constant](#), [Brian Lester](#), [Rami Al-Rfou](#)

**U.S. Patent Application**, 17/863,840

[Task Augmentation and Self-training for Improved Few-shot Learning](#)

[Thang Luong](#), **Tu Vu\***, [Quoc Le](#), [Grady Simon](#)

**U.S. Patent Application**, 17/826,690